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Machine learning and the renewable energy revolution: Exploring solar and wind energy solutions for a sustainable future including innovations in energy storage

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Abstract

This article evaluates the present global condition of solar and wind energy adoption and explores their benefits and limitations in meeting energy needs. It examines the historical and evolutionary growth of solar and wind energy, global trends in the usage of renewable energy, and upcoming technologies, including floating solar and vertical-axis wind turbines. The importance of smart grid technology and energy storage alternatives for enhancing the effectiveness and dependability of renewable energy is explored. In addition, the role of Electric Vehicles (EVs) in a modern smart grid has been assessed. Furthermore, the economic benefits, and most recent technological developments of solar and wind energy and their environmental and social ramifications. The potential of solar and wind energy to meet the increasing global energy demand and the problems and opportunities facing the renewable energy industry have shown excellent promise. Machine learning applications for solar and wind energy generation are vital for sustainable energy production. Machine learning can help in design, optimization, cost reduction, and, most importantly, in improving the efficacy of solar and wind energy, including advancing energy storage. This assessment is a crucial resource for policymakers, industry leaders, and researchers who aim to make the world cleaner and more sustainable. Ultimately, this review has shown the great potential of solar and wind energy in meeting global energy demands and sustainable goals.

KEYWORDS

energy storage, machine learning, renewable energy, solar energy, wind energy

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1 | INTRODUCTION

As nations contend with the dual challenges of economic and social development at the dawn of the 21st century, the issue of energy has been thrust into the limelight due to growing concerns regarding its consumption, accessibility, and sustainability (Djellouli et al., 2022; Milin et al., 2022). Global energy consumption, a crucial pillar of economic development and social welfare, is increasingly recognized as a significant contributor to the world's environmental problems, particularly climate change (Zhao, Dong, et al., 2022). In recent years, the world has experienced a convergence of energy crises characterized by the overexploitation of non-renewable energy resources and a growing environmental awareness. To resolve these issues, a paradigm shift towards renewable energy, a less damaging alternative to non-renewable energy, has emerged as a global development agenda (Wang & Zhan, 2019). This shift is evident in the accelerating transition of the energy structure towards carbon neutrality, with countries committing to international efforts for low-carbon energy and climate change mitigation. The "World Energy Outlook 2020" by the International Energy Agency emphasizes renewable energy resilience despite the COVID-19 pandemic that led to a global decline in energy demand (Energy, n.d.; Zhao, Dong, et al., 2022). The ongoing war in Ukraine has profoundly impacted the global energy sector, particularly in Europe, leading to significant disruptions in natural gas supplies and escalating energy prices (Cui et al., 2023; Kuzemko et al., 2022). This crisis has accelerated the urgency for energy independence in affected regions, particularly in Europe, prompting a re-evaluation of renewable energy targets and strategies (Rokicki et al., 2023; Van de Graaf, 2023). Countries heavily reliant on Russian gas, such as Germany and Italy, have faced substantial challenges, catalyzing a shift towards more diversified and sustainable energy sources (Balsalobre-Lorente et al., 2023; Radtke & Wurster, 2023). The geopolitical instability has also highlighted the resilience of renewable energy sources, potentially influencing future investment and policy decisions towards accelerating the adoption of solar and wind energy.

According to projections, renewable energy's electricity consumption will increase by two-thirds between 2020 and 2030, accounting for nearly 80% of global electricity demand. This trend indicates the renewable energy sector's rapid growth and immense potential (Bank, 2005). The global transition to renewable energy is only, however, with obstacles. Energy poverty, defined as the absence of access to clean fuels, facilities, and the corresponding energy services,

continues to be a pressing issue. In 2019, a staggering 2.6 billion people lacked access to sanitary cooking facilities, with coal, solid biomass, and kerosene as their primary fuels (Zhao, Dong, et al., 2022). In 2015, the United Nations adopted the Sustainable Development Goals (SDGs) to establish a sustainable future by 2030, focusing on various aspects of human well-being, including access to affordable and clean energy. Solar and wind energy are essential to achieving these goals, as they are two of the most promising renewable energy sources. In addition to mitigating the adverse effects of climate change, they provide clean, sustainable, and increasingly cost-effective solutions for meeting the world's growing energy demands. The complexities of energy poverty in developing nations, characterized by a lack of access to modern energy sources, emphasize the need for individualized solutions (Sy & Mokaddem, 2022). Such efforts align with the 2030 Agenda for Sustainable Development's emphasis on energy access and sustainability (Güney, 2019). As the world continues to face the challenge of climate change, the need for a just transition to a low-carbon economy via well-regulated and balanced solutions has become more apparent. This transition, which centers on energy security, equitable access to energy, and environmental sustainability, has far-reaching implications for global energy conservation and emissions reduction (Hassan et al., 2022).

Despite the considerable increase in the global installed capacity of solar photovoltaics and wind energy, renewable energy sources face inherent challenges, such as intermittency, weather-dependent production, and high initial investment costs (Obaideen et al., 2021). However, the long-term benefits of renewable energy development, such as environmental protection, reduction of greenhouse gas emissions, and creation of new business opportunities, typically outweigh the costs. To attain the seventh Sustainable Development Goal of the United Nations, which emphasizes the need for affordable, sustainable, reliable, and modern energy for all (Dalei et al., 2021), overcoming these obstacles and encouraging the widespread adoption of renewable energy. As the world transitions to sustainable energy systems, integrating renewable energy, smart grid technology, energy storage, and energy security becomes more critical (Maka & Alabid, 2022). To fulfill the world's energy needs sustainably by 2030, the International Energy Agency (IEA) estimates that annual investment in renewable energy must exceed \$1 trillion (Gielen, 2017). Governments, international institutions, and private businesses have implemented various financing mechanisms to encourage such investments, including public-private partnerships, green bonds, and crowdsourcing (Tian, 2018).

Artificial intelligence (AI) has emerged as a transformative force in the renewable energy sector, potentially revolutionizing solar and wind energy solutions by optimizing production, increasing environmental awareness, and enhancing management and distribution (Belu, 2013; Jha et al., 2017). AI applications have proven their ability to improve wind farm management, decision-making, and offshore wind energy production (Mitchell et al., 2022). As a subset of AI, machine learning has been increasingly applied to solar and wind energy systems to optimize their efficacy, lower costs, and improve their predictive capabilities. Machine learning techniques can improve the application and administration of renewable energy sources, making them more viable options for meeting the world's energy needs. In solar energy systems, machine learning algorithms enhance solar panel performance, increase energy forecasting, and optimize energy storage systems. For instance, machine-learning techniques have been used to detect and localize solar panel faults, drastically reducing the time required to identify and rectify faulty cells (Ahan et al., 2021). In addition, machine learning-based solar energy forecasting systems have demonstrated high accuracy (Partheeban et al., 2022). Similarly, machine learning has shown promise in wind energy systems, where it can enhance wind energy forecasting and wind speed predictions (Zhao, Sun, & Wang, 2022). Moreover, machine learning algorithms have optimized wind farm planning, resulting in more efficient design and implementation (Wu et al., 2014). By leveraging the predictive and optimization capabilities of machine learning, these renewable energy sources can be harnessed and managed more effectively, substantially contributing to the achievement of global energy demands and sustainable development objectives.

This review analyses machine learning's role in developing renewable energy, concentrating on solar and wind energy solutions and energy storage innovations. The difficulties and limitations of the current state of renewable energy and AI technology is discussed while emphasizing the sector's AI-driven advancements. The discussion also extends to the potential of solar and wind energy for a sustainable energy future. Ultimately, the purpose of this work is to highlight the intersection of machine learning with solar and wind energy in shaping the future of solar and wind energy solutions and innovations in energy storage, contributing to a better understanding of AI's potential to advance the renewable energy sector and achieve a more sustainable future.

2 | RENEWABLE ENERGY ENVIRONMENT

2.1 | Overview of solar energy

As one of the most abundant and purest renewable energy sources, solar energy has the potential to significantly advance sustainable development goals. Because solar photovoltaics are the most cost-effective renewable energy source, their global deployed capacity has increased rapidly over the past few years (Ahmadi et al., 2018). In 2017, solar and wind energy accounted for more than 50% of the global increase in electricity generation capacity, with solar installations surpassing the combined total of new fossil fuel and nuclear power capacities (Burke et al., 2019).

Utilizing semiconductors to collect, separate, and transport photogenerated electron-hole pairs to electrodes to maximize solar absorption is a technological advance in photovoltaic systems that seeks to improve the efficiency and viability of solar energy (Durganjali et al., 2020). Despite solar energy's potential, its contribution to global energy production is negligible relative to other energy sources (Ahmadi et al., 2018). The future of solar energy depends on government policy, technological innovation, and energy competitiveness. From 2006 to 2016, policy support, technological advancements, and declining prices contributed to a 50% increase in solar electricity production. A revolution in energy is required to attain net-zero emissions by 2050 (Chu & Meisen, 2011).

Various varieties of solar cells have emerged as solar technology has advanced, including thin-film, crystalline silicon, concentrator, and perovskite solar cells (Korniejenko et al., 2021). Perovskite solar cells, for example, have a simpler and less expensive production method than silicon solar panels (Wang et al., 2021). Although concentrated solar power systems such as parabolic troughs, solar towers, and dish/engine power plants use solar radiation to generate electricity, their high costs and lack of adaptability have prevented their pervasive deployment in large-capacity networks (Islam et al., 2018).

Solar energy provides numerous benefits, including the capacity to sustainably satisfy rising electricity demand. Solar panels can provide abundant, low-carbon energy, thereby fostering energy security, decreasing fossil fuel consumption and emissions, and meeting the rising electricity demand. As a result of their environmental impact and energy loss, solar panels are considered hazardous waste (Korniejenko et al., 2021). Solar energy has the potential to provide significant environmental and social benefits, and ongoing research is driving solar power's viability towards a more efficient and sustainable approach. Further research is required to improve the durability and performance of solar cells, and recycling solutions must be developed to resolve the problem of toxic waste. Achieving net-zero emissions by 2050 will necessitate a vast transition to renewable energy, with solar power playing a crucial role in achieving this objective.

2.2 | Overview of wind energy

Wind energy has acquired prominence as a renewable energy source due to the need to reduce carbon emissions and the rising electricity demand. This renewable and sustainable energy source relies predominantly on electromagnetic generators (EMGs) to convert wind energy into electricity. Nevertheless, EMGs have disadvantages, such as high maintenance costs and limited voltage output, which necessitate the development of alternative energy conversion technologies (Ellabban et al., 2014).

Triboelectric nanogenerators (TENGs) have emerged as promising devices that can capture a variety of mechanical energies, including wind energy. TENGs offer benefits such as high voltage, low weight, and simple production, but their output is still modest (Qian & Jing, 2018). Researchers have created hybrid nanogenerators that combine TENG and EMG technologies for self-powered wind speed sensing and fuelling commercial solar cells (Guo et al., 2019).

Nevertheless, these hybrid nanogenerators encounter obstacles such as high wind speed cut-ins and complex designs, necessitating innovative designs that reduce friction and boost output performance.

Recent innovations, such as a pinwheel that generates electricity continuously by combining triboelectrification and electromagnetic induction, demonstrate the potential to capture small quantities of wind energy (Guo et al., 2019). Additional research and development in this area may result in the creation of more efficient and cost-effective wind energy harvesting systems. Wind energy has the potential to generate pure, non-polluting electricity on a global scale and is a rapidly expanding energy source. While EMG-based wind turbines continue to be the most prevalent method for converting wind energy into electricity, some limitations must be addressed. Researchers have developed nanogenerators to serve as self-powered wind speed sensors or power commercial solar cells (Zhao, Wu, et al., 2022). TENGs represent a viable energy conversion technology for wind energy collection. Novel hybrid TENG designs with low friction and high output performance are still required, and recent developments in pinwheels incorporating triboelectrification and electromagnetic induction demonstrate the potential for more efficient and cost-effective wind energy harvesting devices.

2.3 | Current status and emerging renewable energy technologies

The viability of renewable energy as a sustainable recovery strategy post-COVID-19 has been highlighted by various studies (Hosseini, 2020). However, incorporating renewable energy sources, especially solar and wind, into existing systems encounters obstacles, such as fluctuating petrol and oil prices, economic factors, and policy barriers. The key to circumventing these obstacles is the development of innovative AI technologies and energy storage strategies and the harmonization of legislation and economic incentives to promote sustainable energy transitions.

Countries like India and Indonesia have made substantial progress in adopting renewable energy by implementing targeted policies such as the Jawaharlal Nehru National Solar Mission and the Solar Energy Corporation of India (SECI). Nonetheless, these nations continue to face legislative, financial, and infrastructural obstacles that can be addressed through feed-in tariffs, subsidies, tax credits, and regulatory adjustments (Sahu, 2018). In addition, developing enhanced transmission and distribution infrastructure, real-time power markets, and frequency-regulating systems can facilitate the integration of renewable energy sources.

Floating solar PV systems offer advantages over land-based systems, including reduced land use and enhanced efficiency due to the cooling effect of water (Mamatha & Kulkarni, 2022; Manoj Kumar et al., 2022). The assertion is that the enhancement of FPV plants' capacity, ranging from 52 kW to 2 MW, can significantly reduce the levelized cost of energy (LCOE) by up to 85%, with further research, developments, and advancements in technology and materials potentially making FPV technology more affordable in the future (Gorjian et al., 2021).

Utilizing AI to optimize the performance of these systems presents opportunities for further development. Research in wind turbine technology, such as blade design, vertical-axis wind turbines, and accurate wind speed forecasting, has the potential to improve wind farm performance and environmental compatibility (Hui et al., 2018; Sitharthan et al., 2018; Staid & Guikema, 2015; Vargas et al., 2019).

In the renewable energy industry, AI technologies play a crucial role in addressing the challenges posed by the COVID-19 pandemic. By integrating AI into distant control systems and decentralized energy production, renewable energy can support post-pandemic recovery and contribute to developing a low-carbon economy (Hosseini, 2020). Governments and policymakers must prioritize strategic actions that support renewable energy initiatives and ensure a sustainable energy transition.

In conclusion, integrating renewable energy sources, particularly solar and wind, into existing systems is essential for a sustainable future. As nations recover from the COVID-19 pandemic, AI-driven innovations in renewable energy technologies, such as floating solar PV systems and advanced wind turbines, have the potential to significantly accelerate the transition to pure, sustainable energy sources. By investing in research and development, nurturing supportive policies, and promoting collaboration between governments, industry leaders, and academia, we can advance a greener future and reduce dependence on fossil fuels.

2.4 | Global trends in the adoption of renewable energy

Due to limitations in availability and acceptance, reliance on traditional renewable energy sources and fossil fuels, and environmental, social, and economic repercussions of energy transitions, adopting renewable energy sources in developing countries is a crucial issue. While biomass and hydropower have been utilized in developing nations for quite some time, solar and wind energy are gaining popularity (Vane-gas Cantarero, 2020). Despite the limited use of contemporary renewable energy in these nations, non-OECD countries accounted for 72% of the world's 13.5% in 2017. The Global South, which has the lowest proportion of renewable energy, continues to rely heavily on traditional biomass. This substantial reliance on solid biofuels raises concerns regarding social equality, energy efficiency, environmental and health consequences, and land use (Solarin et al., 2018).

Energy systems are progressively incorporating renewable energy sources to address environmental concerns.

Since 2015, developing nations have invested more in modern renewable energy than developed nations, with solar photovoltaics and onshore wind becoming the dominant energy sources in emerging nations due to their decreasing costs (Vanegas Cantarero, 2020). While floating solar PV systems are emerging as a viable alternative in areas with limited land, they pose unique challenges, such as the potential impact on aquatic ecosystems and higher maintenance costs due to moisture and corrosion (Gorjian et al., 2021; Nobre et al., 2023). Additionally, the stability and durability of floating

platforms in diverse weather conditions remain concerns, potentially affecting these systems' long-term viability and energy yield (Huang et al., 2023; Kumar et al., 2021). However, many developing nations continue to use traditional renewable energy sources in addition to fossil fuels like coal, oil, and natural gas. In several developing nations, notably rural regions with limited access to electricity and clean fuels, energy poverty remains a significant problem.

The transition to renewable energy in developing countries faces numerous obstacles, such as size and reliability constraints in power systems that lead to technical and non-technical losses, adverse market fluctuations, tax collection, and energy subsidies, which can contribute to the financial instability of the power sector (Qiu et al., 2022; Saini et al., 2023). Moreover, the excessive reliance on automobiles powered by fossil fuels raises concerns regarding energy efficiency (Fuldauer et al., 2022).

In conclusion, despite substantial investments in renewable energy, developing countries continue to rely on traditional renewable energy sources and fossil fuels for their energy systems. Adopting renewable energy sources confronts many obstacles, including energy poverty, size and reliability limitations in power systems, adverse market fluctuations, tax collection, energy subsidies, and excessive reliance on fossil-fuelled vehicles. To facilitate a successful transition to renewable energy in these nations, it is essential to address energy scarcity and improve data accessibility.

3 | INNOVATIVE APPROACHES FOR INTEGRATING SOLAR AND WIND ENERGY

3.1 | Smart grid

In recent decades, renewable energy sources have transformed electricity distribution, posing new challenges to grid systems, such as variable energy output and remote power plant locations (Rangel-Martinez et al., 2021). Smart grids, which connect electric infrastructure, storage facilities, and consumers, have emerged to assure energy efficiency, reduce greenhouse gas emissions, and cost-effectively regulate power supply (Fatima et al., 2020; Rangel-Martinez et al., 2021). Smart grids have incorporated machine learning (ML) to predict price fluctuations, capacity projections, and fault detection.

Hossain et al. (2019) investigate the application of big data and machine learning within the Internet of Things (IoT)-integrated smart grid (SG) system. Machine learning proves indispensable for processing massive data volumes, providing predictive capabilities, and facilitating informed decision-making, thereby optimizing the operation of the grid. In addition to load forecasting and fault detection, other applications include user preference analysis (Hossain et al., 2019).

Ahmad et al. (2022) study employing machine learning (ML) techniques to optimize decision-making and energy distribution in smart grids, highlighting two areas: core energy technologies and energy distribution utilities. The research delves into multiple ML models such as Supervised, Unsupervised, Reinforcement learning, Bayesian methods, Probabilistic ML models, Logistic Regression, and

Deep Neural Networks. These models demonstrate their effectiveness in data and load forecasting, optimal energy distribution, and managing uncertainties tied to renewable energy integration. The study amalgamates data-driven probabilistic ML techniques with real-time applications to control dynamic energy systems. It estimates potential savings of \$237 billion to \$813 billion for the energy industry with smart ML automation.

By incorporating ML and artificial intelligence, smart grids can intelligently respond to sudden shifts in power supply and demand and forecast energy needs and potential equipment malfunctions (Azad et al., 2019). Traditional fuels and batteries cannot store enough electricity for urban areas, making load forecasting critical. ML algorithms, such as ensemble learning, have shown promising results, frequently outperforming conventional forecasting models (Rangel-Martinez et al., 2021). For instance, the work by (Naganathan et al., 2016) further supports the versatility of ML, deploying semi-supervised learning to model energy consumption for buildings and at an industrial scale, respectively. They effectively tackled the challenge of scarce labeled data, a common issue in ML. Other ML applications include semi-supervised learning for modeling energy consumption in buildings and dynamic security assessment (DSA) for assessing the resilience of a system.

Smart grids are essential for sustainable future cities because they integrate renewable and non-renewable energy sources to reduce environmental impacts (Saidani Neffati et al., 2021). They facilitate real-time solutions and offer new amenities and economic value to residents of smart homes. The concept of smart cities addresses future urbanization challenges, with countries such as India investing heavily in smart grid initiatives. The inability of traditional models, such as Integrated Assessment Models (IAMs) and Price-Induced Market Equilibrium Systems (PRIMES), to provide precise hourly or annual supply and demand projections hinders the integration of solar energy into the grid. Smart grid architectures that prioritize energy storage systems (ESS) can be implemented to improve asset utilization, system reliability, and demand responsiveness (Worighi et al., 2019). Several essential characteristics of smart grids include self-healing, interactivity, data security, and compatibility with ESS (Xue-Song et al., 2010). Integrating wind farms into utility systems while preserving electricity quality and system security is crucial (Colak et al., 2015).

Integrating smart grid technologies is crucial for addressing the challenges posed by expanding renewable energy sources. Smart grids offer numerous advantages, such as energy efficiency, greenhouse gas reduction, and cost-effective administration of the power supply (Kabeyi & Olanrewaju, 2022; Moreno Escobar et al., 2021; Moretti et al., 2017; Popkova et al., 2023). In load forecasting, energy consumption modeling, and DSA, applying ML algorithms has yielded encouraging results. Smart grids are essential to developing sustainable smart cities by integrating renewable and non-renewable energy sources. Governments around the globe, including India, have recognized the significance of smart grids in overcoming scientific, economic, and regulatory barriers and have made substantial investments in their development. Integration of smart grid technologies is crucial for the long-term sustainability of our energy systems and urban environments.

3.1.1 | The role of electric vehicles (EVs) in a modern smart grid

This section explores electric vehicle (EV) charging and discharging strategies, focusing on Vehicle-to-Grid (V2G) technology and its impact on energy distribution networks. It examines the pivotal role of V2G in enhancing grid functionality, including aspects like active power regulation, reactive power support, load balancing, and harmonic filtering. The study also acknowledges the challenges accompanying these advancements, such as potential battery life reduction, communication complexities, and necessary changes in distribution network infrastructure. Key topics discussed include the influence of EV penetration levels, charging profiles, and the implications of various controlled charging and discharging methods on power quality and network losses.

The article by Mastoi et al. (2023) presents a detailed examination of various EV charging and discharging strategies, particularly focusing on V2G technology and its impacts on energy distribution networks. It highlights the benefits of V2G, such as active power regulation, reactive power support, load balancing, and current harmonic filtering. However, challenges like reduced battery life, communication overhead between EVs and grids, and changes in distribution network infrastructure are also acknowledged. Key aspects discussed include the effects of EV penetration levels, charging profiles, and controlled charging and discharging on power quality and loss. The study comprehensively analyzes methods like controlled and uncontrolled charging–discharging, delayed charging–discharging, indirectly controlled discharging, bidirectional charging–discharging, and intelligent scheduling. It is noted that Li-ion batteries can be recharged 2000–4000 times and cost \$200–\$500 per kWh. The degradation cost of batteries at 80% discharge depth is approximately \$130 per MWh for a 300-kWh investment. The paper also delves into the technicalities of unidirectional and bidirectional power flow, discussing the merits and challenges of each. Various charging and discharging strategies are explored, including the impacts on the grid, utility, voltage regulation, and power quality. It is revealed that smart metering and charging can increase self-consumption by 45%. Finally, the article proposes future research directions and recommendations for V2G technical development, emphasizing the need for enhanced V2G modeling, capacity forecasting, and the consideration of social aspects in electrified transportation systems.

The work by Habib et al. (2015) provides a comprehensive review of V2G technology and its integration with various EV charging strategies, assessing their effects on power distribution networks (DNS). The work highlights that V2G-equipped vehicles offer active and reactive power regulation, load balancing, and harmonics filtering functionalities. Nonetheless, V2G faces challenges such as battery degradation, communication complexities, and infrastructural modifications in DNSs. The study emphasizes the importance of charging profiles and EV penetration levels on DNSs, pointing out that a single EV can add about 560–910 W to system load at peak times, influenced by the charger's size and the charging strategy. It distinguishes between coordinated/controlled and uncoordinated/uncontrolled

charging, delayed charging, and off-peak charging strategies. Uncoordinated charging, lacking intelligent scheduling, significantly impacts DNSs, often leading to increased load on distribution elements like transformers, thereby reducing grid reliability. Conversely, coordinated charging, which optimizes charging times during low-demand periods, is presented as an effective solution to mitigate these issues. It is noted that smart charging and discharging can reduce daily electricity costs, voltage deviations, and transformer and line current surges, thereby enhancing the DNS's stability, efficiency, and reliability. The paper underscores that coordinated charging benefits EV owners and grid operators, especially with higher EV penetration. The paper also discusses the potential of V2G technology to improve grid reliability, efficiency, loss mitigation, and stability. Smart metering and intelligent grid communication are deemed essential for effective V2G operation. However, the economic viability of V2G is contingent upon charging/discharging strategies and vehicle aggregation.

In addressing the integration of EVs into modern smart grids, the work by Ahmadi et al. (2019) elucidates the multifaceted role EVs play in contemporary electrical networks. EVs contribute significantly to environmental sustainability by reducing emissions when replacing fossil-fuel vehicles. However, their widespread adoption introduces challenges, particularly in power quality (PQ) within electrical grids, due to their grid-to-vehicle (G2V) and V2G capabilities. The rapid increase in EVs, as evidenced by global sales surpassing 750,000 units in 2016, necessitates a closer examination of their impact on power systems. This includes the potential of V2G technologies to facilitate power exchange between EVs and the utility grid. However, the integration of V2Gs into existing power grids remains underdeveloped, requiring further research and optimization. Concerns also arise regarding the grid's capacity to handle the charging and discharging of EV batteries, especially considering the need for improved charging technologies and smart metering. The development of EVs is closely tied to the establishment of private and public charging infrastructures. With a 72% growth in publicly accessible charging points since 2015, government interventions are crucial for supporting this expansion. Additionally, advancements in battery technologies, including those offering higher power density and larger capacities, are essential for the sustainable deployment of EVs. Overall, the successful integration of EVs into smart grids depends on the collaborative efforts of governments, power utilities, and EV owners. It also hinges on developing cost-effective, multifunctional EV chargers capable of providing ancillary services, such as voltage regulation and harmonic reduction, in both G2V and V2G modes. This necessitates a multi-layer, communication-enhanced infrastructure for coordinating between system operators, EV aggregators, and end-users.

In conclusion, this section underscores the transformative role of V2G technology in conjunction with smart EV charging strategies for power distribution networks. It highlights that well-coordinated charging methods, particularly those that optimize charging times during low-demand periods, offer substantial benefits for grid stability and efficiency. The integration of V2G technology with intelligent grid communication systems is crucial for the successful operation of this synergy. Future research directions suggest a need for enhanced

modeling of V2G systems, improved forecasting of capacity requirements, and a consideration of social factors in electrified transportation systems. Additionally, further exploration into the economic viability of V2G, particularly concerning battery degradation costs and charging/discharging strategies, remains paramount. Advancements in this field promise a more resilient, efficient, and sustainable power grid harmoniously integrated with the rising tide of electric vehicles.

3.2 | Overview of energy storage for renewable energy

Integrating renewable energy sources, particularly solar and wind energy, into power systems depends on energy storage. Figure 1 depicts the basic schematic representation of the integration of energy storage. This section examines recent developments in energy storage technologies and artificial intelligence's role in optimizing their implementation and operation for a sustainable future. The intermittent nature of solar and wind energy poses a challenge to attaining a consistent power supply, making energy storage essential. Integrating micro-compressed air energy storage (micro-CAES) into photovoltaic and wind energy systems enables effective demand shifting (Arnaoutakis et al., 2023). Algorithms based on AI can model and optimize energy distribution based on predetermined criteria, resulting in more efficient use of renewable resources and reduced energy losses.

Battery energy storage systems (BESSs) are another prevalent solution for addressing the intermittent nature of renewable energy. Several factors, including operating temperature, depth of discharge, and charging/discharging currents, influence the lifespan of batteries in BESSs (Khalid Mehmood et al., 2017). AI can optimize the location, size, and operation of battery energy storage systems (BESSs), considering factors influencing battery lifespan and uncertainties in solar and wind power generation. This strategy reduces losses, enhances voltage regulation, and lengthens battery life.

In the work by Mohamad et al. (2021), an innovative combinatorial optimization framework is developed to maximize the effectiveness of Battery Energy Storage Systems (BESSs) in a solar-integrated

power network. The framework comprises two parts: the first optimizes BESS placement within the network while ensuring maximum connectivity to all busses, and the second determines the capacity level of each BESS, factoring in power flow constraints. The framework demonstrates notable efficiency gains when applied to the IEEE 24-bus RTN, which includes 38 transmission lines and multiple generator and load busses. Key findings include identifying three equitable BESS placement solutions involving eight mandatory busses. Even before capacity distribution optimization, these optimized placements show an average reduction in Energy Storage Capacity (ESEC) by 8.85% compared to conventional deployment strategies. The most effective solution, Solution 2, significantly optimizes BESS capacity distribution, primarily concentrating it in regions with lower line capacity, achieving an average of 82% concentration. This strategic placement and capacity allocation further reduced ESEC by approximately 27% across all solutions, demonstrating the framework's effectiveness over traditional methods. Sensitivity analyses further underscore the framework's robustness. Adjustments in line ratings and BESS capacities reveal that ESEC decreases with increased BESS capacity and improves line ratings, highlighting the impact of network constraints on solar power utilization. Additionally, exploring varying C-rates and reliability levels of BESSs shows an inverse relationship between ESEC and the Expected Solar Energy Stored (ESES), adding another dimension to the framework's applicability.

Roy et al. (2020) presents a novel hybrid energy storage system (HESS) for a wind-solar hybrid power system (WSHPS) simulated in MATLAB/Simulink. Using a combination of bidirectional power converters, a three-level T-type inverter, and an LCL filter, the research demonstrates a constant power dispatch to the utility grid, with fast-changing power components charged/discharged by short-term energy storage system (SESS) and slow-changing components by battery energy storage system (BESS). The study used two schemes for generating the dispatchable power reference case and load following case, with promising results in both. HESS demonstrated optimum performance at 42% depth of discharge (DOD) for BESS, yielding the lowest cost of about 9.85 €/kWh, while SESS exhibited minimum cost at 100% DOD, approximately 17.2 €/kWh. The simulation results

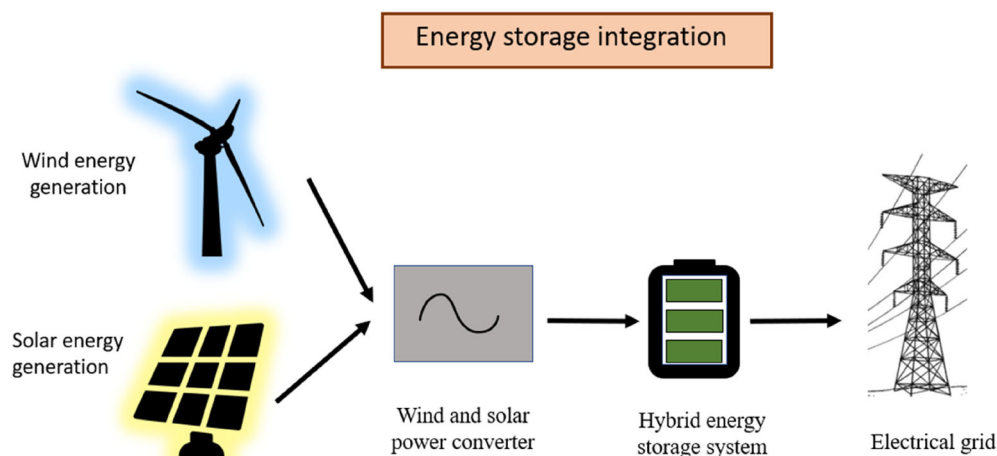


FIGURE 1 Schematic of an energy storage integration.

showed that HESS can be more cost-effective than battery or super-capacitor-only operations. The paper further explores the impact of different parameters on the cost of HESS and offers a comprehensive economic comparison. An optimal Low Pass Filter (LPF) time constant of 62 s was determined for HESS cost using a Particle Swarm Optimization (PSO) strategy (Roy et al., 2020).

Seasonal energy storage is essential for increasing the penetration of wind and solar photovoltaic energy. For grid-integrated seasonal storage techno-economic analyses, AI-driven modeling techniques can help identify the most cost-effective alternatives for storage technologies such as pumped hydro, compressed air, and hydrogen (Guerra et al., 2020). AI-assisted analysis can aid in calculating storage operation profiles, estimating the cost-effectiveness of storage devices, and accounting for avoided power system costs.

The study by Pattanaik et al. (2021) focuses on the integration of a Permanent Magnet Induction Generator (PMIG)-based wind system with a Dynamic Voltage Restorer (DVR) and an Energy Storage System (ESS) for backup power in a novel manner. The goal is to regulate fluctuations in wind power and counteract grid voltage disturbances. The effectiveness of a bidirectional converter in controlling power flow in both directions, from the source to the battery and the battery to the voltage source inverter (VSI), thereby compensating for grid voltage disturbances, is a key finding. An intriguing aspect of the methodology was using MATLAB/Simulink to validate the results. Important outcomes of the simulation included a compensated sag voltage of the grid terminal voltage, a regulated DC link voltage, and an efficient bidirectional power flow in the energy storage battery system. The study demonstrates the potential of DVR-based energy storage systems for wind energy conversion and emphasizes the vital role of energy storage in grid stability maintenance. It highlights the importance of the DVR system in compensating for under voltage while maintaining grid stability, thereby highlighting the significance of energy storage in renewable energy systems (Pattanaik et al., 2021).

Battery energy storage systems (BESS) have been utilized to resolve power quality issues in wind turbine systems, such as power stabilization, power factor correction, voltage control, frequency control, and time shift (Pontes et al., 2023). To demonstrate BESS's efficacy in various applications, AI-driven tools, such as HOMER Pro, can be used for system simulations and real-world data analysis. The various applications of battery energy storage systems and hybrid energy storage systems in wind turbine systems are shown in Figure 2.

In conclusion, artificial intelligence contributes significantly to developing solar and wind energy systems and energy storage solutions. AI-driven optimization and modeling techniques can enhance energy storage systems' efficacy, cost-effectiveness, and dependability, paving the way for a more sustainable and resilient energy future.

3.3 | Renewable energy systems integration, technology advancement, and potential of solar and wind energy

Integrating renewable energy sources such as solar and wind into conventional energy systems is crucial for mitigating the environmental

impact of fossil fuel-based power facilities and promoting sustainable development. However, successful integration is hindered by power quality, availability, generation instability, forecasting, facility location, and financial constraints (Ayadi et al., 2020; Østergaard et al., 2020). Improving the efficacy of renewable energy systems necessitates accurate wind and solar resource forecasting. Staid and Guikema (2015) and Vargas et al. (2019) report that the development and implementation of advanced forecasting models and algorithms, as well as the incorporation of artificial intelligence and machine learning techniques, can substantially improve the accuracy of these forecasts. In addition, managing reserves and integrating energy storage technologies can assist in overcoming the challenges posed by the intermittent nature of solar and wind power generation (Hoang et al., 2021).

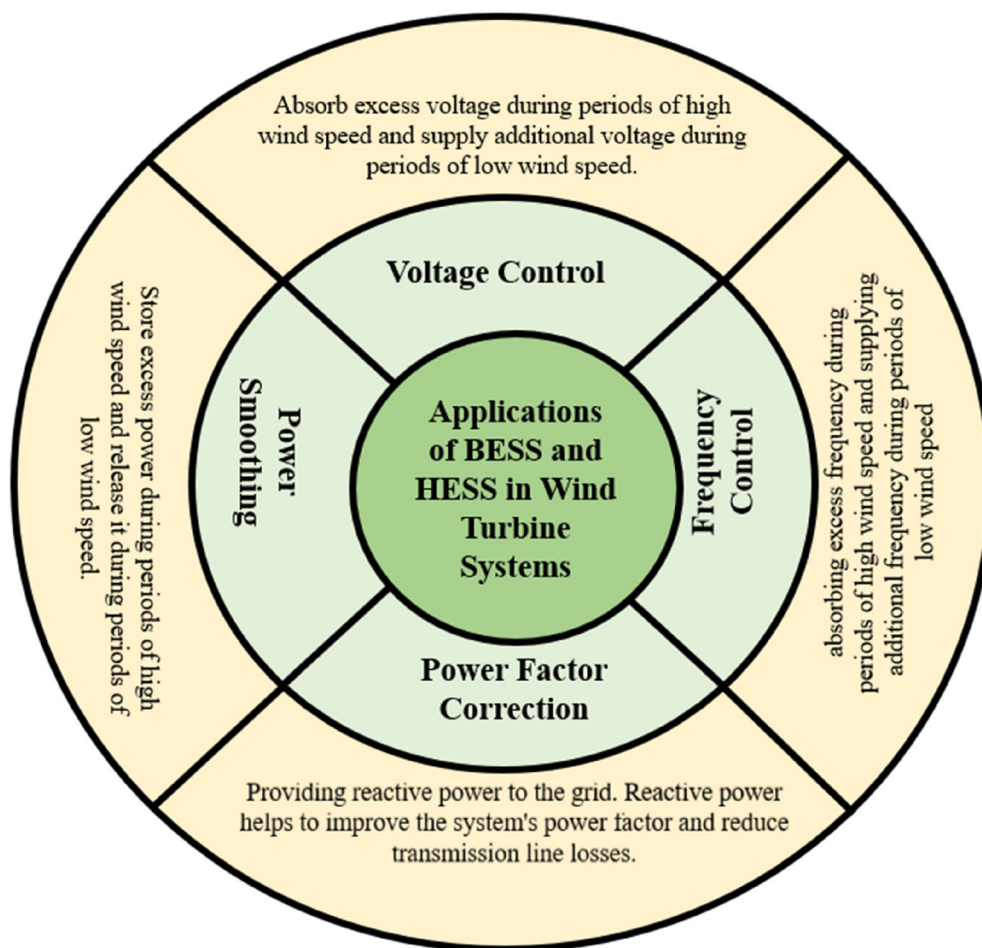
Optimizing the integration of renewable energy systems in smart communities is crucial to sustainable urban development. The focus of research should be on the roles, objectives, design principles, and instruments of integrated renewable energy systems (IRES) and the opportunities and challenges they present (Maheshwari & Ramakumar, 2017). By analyzing factors such as management, system performance, interconnection, and interactions between various renewable energy systems, researchers can identify best practices for large-scale deployment of IRES (Hoang et al., 2021). Wind energy is a promising renewable energy source that has the potential to reduce greenhouse gas emissions and our dependence on fossil fuels. Innovations in wind turbine design, such as horizontal-axis and vertical-axis wind turbines (HAWTs and VAWTs), have substantially enhanced the efficacy and stability of wind energy systems (Shoab et al., 2019). In addition, developments in blade design, materials, and control systems have contributed to the expansion of the wind energy industry (Hui et al., 2018; Sitharthan et al., 2018).

In transitioning to a sustainable energy future, solar energy, notably photovoltaic (PV) technology, is another crucial component. Despite the obstacles of high installation costs and intermittent power generation, solar energy has the potential to become the primary source of renewable electricity (Kabir et al., 2018; Kannan & Vakeesan, 2016). The development of more efficient solar panels, innovative energy storage solutions, and supportive policies can assist in overcoming these barriers and accelerating the global adoption of solar power (Kumar et al., 2016; Kumar et al., 2021; Mamatha & Kulkarni, 2022).

A balance between renewable and non-renewable energy sources is vital for achieving economic growth while minimizing climate damage (Bird et al., 2016). Reduce greenhouse gas emissions and replace non-renewable energy sources with renewable energy. Wind energy, in particular, is increasing rapidly and provides endless eco-friendly electricity (Shoab et al., 2019). Horizontal-axis and vertical-axis wind turbines are the most prevalent forms of wind turbines, with category three wind turbines suitable for small-acreage sites (Stathopoulos et al., 2018). Notwithstanding challenges, building-integrated wind energy systems offer numerous benefits, including land conservation and improved efficiency (Stathopoulos et al., 2018).

Production of offshore wind energy involves problems, although floating turbines enable production in deeper seas (Oh et al., 2018). Thanks to feed-in tariffs and state-level support, China, India, and

FIGURE 2 The applications of battery energy storage systems and Hybrid energy storage system in wind turbine systems.



Japan have expanded their wind energy sectors. At the same time, Germany has the largest wind market in the European Union and mandates a wind electricity mix (Ho, 2016). Under Contracts for Difference, the United Kingdom is increasing offshore wind capacity and has a renewable energy target of 15% by 2020. Renewable energy pricing is determined by technology, natural resources, and economies of scale, with crowdfunding, YieldCos, and sustainable lifestyles serving as market drivers (Lam & Law, 2018). Renewable Portfolio Standards, feed-in tariffs, tax incentives, and subsidies can encourage renewable energy production. To increase the adoption of renewable energy technology, market participant diversity and project financing must be widened (Lam & Law, 2018). A move to renewable energy is vital to counteract climate change and achieve sustainable economic growth. Wind energy, in particular, has immense potential and can be used through various technological advances and financial incentives. Education and awareness activities are also required to encourage sustainable lifestyles and the usage of renewable energy.

In conclusion, integrating solar and wind energy into existing power systems is essential for a sustainable energy future. We can promote the widespread adoption of these sustainable energy sources by addressing the challenges associated with renewable energy integration using advanced forecasting techniques, energy storage technologies, and supportive policies. In addition, continued investment in research and development can spur additional innovation in

renewable energy technologies and help realize their maximum potential in combating climate change and securing a greener future for our planet.

4 | APPLICATIONS OF ARTIFICIAL INTELLIGENCE IN SOLAR AND WIND ENERGY GENERATION

There has been various application and potential applications of AI in solar and wind energy. This section reviews various recent advancement and future potential applications of AI in this area.

4.1 | Case studies on the applications and innovations of AI for solar energy generation and technology with a focus on machine learning applications

The increasing demand for energy in the residential and commercial sectors has highlighted the necessity of embracing renewable energy technologies. Incorporating artificial intelligence (AI), deep learning, and the Internet of Things (IoT) has significantly improved solar energy generation and technology, which is a promising area. This

convergence of innovations has resulted in significant advances in solar panel fault detection, energy forecasting, solar energy optimization, and energy storage system optimization, among others. Numerous studies have shown the great potential applications of machine learning in the context of solar energy, as depicted in Table 1.

With their development of an End-to-End Fault Detection System, Ahan et al. (2021), who discussed the intersection of AI and solar panel fault detection, made significant progress. Using Electroluminescence (EL) Imaging, they detected and localized issues with solar panels, thereby enhancing performance and reducing warranty and service costs. Their system's high level of cell-level accuracy and low number of false negatives significantly reduced the amount of time required to identify and correct problematic cells.

Partheeban et al. (2022) implemented an IoT and deep learning-based infrastructure for energy forecasting. They demonstrated that AI-based prediction systems, such as ANN, FLC, and adaptive neural fuzzy inference systems, efficiently tackle control and prediction issues. Similarly, Said and Alanazi (2022) proposed integrating solar energy into intelligent networks. In energy forecasting, their S2S LSTM neural network outperformed the baseline, indicating the potential of deep learning models in this domain.

Using AI to optimize solar energy has also produced intriguing results. Ali et al. (2021) examined a solar tracking technology that improves the performance of photovoltaic panels. Their findings demonstrated that optimizing artificial intelligence could improve solar tracking systems. In a related study, Meena et al. (2022) utilized AI algorithms to improve the performance and reduce the cost of photovoltaic (PV) systems, indicating that AI-based solutions may be able to reduce the environmental impact of PV power systems.

AI has proven indispensable in optimizing energy storage systems (ESS), which are essential for power system stability and the use of renewable energy. Abdalla et al. (2021) investigated the purpose, design, and optimization of ESS in power systems. While AI techniques have expanded the practical application of ESS in engineering applications, additional research is still needed into AI's potential in multi-objective control of power system optimization.

The combination of AI, IoT, and deep learning with solar energy systems heralds a revolution in the renewable energy industry, as shown by the various research studies in Table 1. AI algorithms have improved the performance and cost-effectiveness of photovoltaic systems, deep learning models have accurately predicted solar energy generation, and the Internet of Things has accelerated the detection and repair of defective solar cells. These technologies have the potential to promote the development of sustainable energy systems in a significant way. There are four main applications of AI to solar energy, which are depicted in the schematic representation in Figure 3.

4.2 | Case studies on the applications of AI in wind energy generation and technology with a focus on machine learning

The current research on the applications of AI in wind energy addresses topics such as wind energy conversion system control, wind

energy forecasting, wind speed prediction, wind farm planning, and damage identification, as shown in Table 2. Integrating AI in these areas aims to enhance efficiency and reduce costs, making it an essential component of the renewable energy sector. Wind Energy Conversion System Control: Researchers have investigated the efficiency of control systems in wind energy conversion systems (WECS) based on Doubly Fed Induction Generators (DFIG). Zohra et al. (2018) and Arama et al. (2019) both explored the performance of Proportional-Integral (PI) and neural network (NN) regulators in managing the wind power conversion system.

Various research has highlighted the potential for AI technologies to significantly improve wind energy forecasts and wind speed predictions, leading to increased efficiency and cost reduction. For instance, in Wind Farm Planning, AI has been applied for optimization, as demonstrated by Wu et al. (2014), who used AI algorithms to optimize offshore wind farm wind turbine and internal line connections. Their findings suggest that AI-driven planning is more efficient and effective than manual planning, providing a valuable tool for optimizing the design and implementation of wind farms. Furthermore, AI has also been used in damage identification, as demonstrated by Movsessian et al. (2021), who developed an ANN-based method for robust damage identification in a vibration-based structural health monitoring framework. Despite environmental and operational variabilities, their approach demonstrated improved damage detectability across all damage scenarios.

Furthermore, in the context of AI Integration and Hybrid Renewable Energy Systems, researchers have also examined the integration of AI with energy storage systems (ESS) and hybrid renewable energy systems. Abdalla et al. (2021) studied power system ESS functions, classifications, design optimization methods, and applications, suggesting that integrating AI with ESS could enable multi-objective power system optimization control. Zahraee et al. (2016) reviewed AI-based hybrid renewable energy system optimization and scaling research, concluding that AI algorithms are more accurate and converge faster than traditional methods. These case studies underscore the enormous potential of AI to improve wind energy generation and technology, especially in its four main areas of application, as depicted in Figure 4. By integrating AI into various aspects of wind energy, efficiency, accuracy, and cost-effectiveness can be enhanced, playing a critical role in achieving global sustainable development goals and addressing climate change challenges. Future research should focus on further exploring and integrating AI in the wind energy sector to maximize its benefits. The summary of the main process steps of the AI innovations in energy forecasting is shown in Figure 5.

5 | EVALUATION OF SOLAR AND WIND ENERGY

5.1 | Economic feasibility of solar and wind energy

The economic viability of producing renewable energy, such as solar and wind power, has been discussed recently. In 2016, solar energy accounted for only 3.5% of Europe's total energy production, while

TABLE 1 The various application of machine learning to solar energy.

Machine learning technique	Application	Dataset	Performance metrics	Main results	Reference(s)
Hybrid multistep CNN-stacked LSTM	Forecasting solar energy variables (Irradiance and POA)	- Solar irradiance data from Sweihan PV independent power project - POA data from Sweihan PV independent power project, UAE	- Mean absolute error (MAE) - Coefficient of determination (R^2) - Mean squared error (MSE) - Root mean square error (RMSE) - Mean absolute percentage error (MAPE) - Normalized root mean squared error (NRMSE)	- The proposed multistep CNN stacked LSTM model outperformed CNN and -LSTM models in both solar irradiance and POA irradiance prediction. - The model also showed better performance compared to other machine learning techniques like linear regression (LR), support vector regression (SVR), and artificial neural networks (ANN). - The proposed architecture demonstrated robustness against noise-injected data sets.	(Elizabeth Michael et al., 2022)
Least-square support vector machine (LS-SVM)	Short-term solar power prediction (SPP)	- National solar radiation database (NSRDB) - Data from Denver, Seattle, and Miami	- Mean absolute error (MAE) - Mean absolute percentage error (MAPE)	- The proposed LS-SVM-based model significantly outperforms the AR model and achieves better results than the RBFNN-based model in terms of prediction accuracy. - Normalization of solar radiations with the transmissivity produces smaller prediction errors than the sigmoid function normalization. - Including additional meteorological variables, especially sky cover, improves the accuracy of solar power prediction.	(Zeng & Qiao, 2013)
Artificial neural networks (ANN) and gradient boosting regression trees (GBRT)	Forecasting solar energy production in four cities using machine learning.	Preliminary information from four distinct locations around the country. Data divided into three seasons: summer, winter, and rainy.	- Root mean square error (RSME) - Normalized root mean square error (nRSMSE)	- ANN and GBRT were found to have the best performance compared to other models (KNN and lasso regression). - The methodology improved results by considering data for a year and dividing it into three seasons, adopting data in peak hours, and performing different models based on climate change.	(Sivakumar et al., 2022)

(Continues)

TABLE 1 (Continued)

Machine learning technique	Application	Dataset	Performance metrics	Main results	Reference(s)
CNN-LSTM-transformer model	Solar energy production forecasting to facilitate integration of solar energy into grids	Fingrid open dataset	Root mean square error (RMSE), mean absolute error (MAE)	- The proposed CNN-LSTM-transformer model achieved the highest accuracy compared to existing models and other combinations. - The model had the lowest RMSE and MAE values of 0.344 and 0.393, respectively. - The results demonstrated the efficiency of the proposed model in solar energy production forecasting, outperforming current state-of-the-art forecasting models.	(Al-Ali et al., 2023)
Deep neural network (DNN) utilizing cloud motion analysis and machine learning.	Intra-day solar irradiance forecasting to improve the controllability of power plants.	Real-world measurements and simulations.	- Root Mean Square Error (RMSE) - Mean Absolute Error - Precision - Recall - F-measure.	- The DNN model outperforms the persistent algorithm across all cases, achieving more accurate and reliable predictions of incoming solar irradiance. - The model is considered state-of-the-art, even without real-world data, as it performs better than current NWP forecasts for the same time horizon. Actual data is more useful for short-term predictions, while weather information is more beneficial for long-term predictions.	(Dhanalakshmi et al., 2022)
Hybrid model integrating machine learning and statistical approaches	Predicting future solar energy generation in large-scale renewable energy plants	- Time series dataset with a total of 10 MW capacity using polycrystalline solar panels (Poly-Si) and thin-film solar cells (TFSC) - Data with a five-minute resolution	Root mean square error (RMSE)	- The proposed hybrid model outperformed standard individual models and a model using only machine learning. - The hybrid model utilizing both machine learning and statistics showed improved accuracy and performance.	(Vennila et al., 2022)
Support vector regression (SVR)	Estimation of technical potential of rooftop solar PV electricity production in urban areas of Switzerland	Urban areas in 1901 communes in Switzerland	- RMSE (Root mean square error) - RRMSE (Relative root mean square error)	The study found that the annual PV electricity production in Switzerland for urban areas in the 1901 communes is 17.86 TW h.	(Assouline et al., 2017)

TABLE 1 (Continued)

Machine learning technique	Application	Dataset	Performance metrics	Main results	Reference(s)
Reinforced learning	Improving solar energy harvesting efficiency through nanomaterial-based solar panels and tracking the Sun's path using machine learning.	Generated by the Sun path calculator app, consisting of 13,140 data elements collected over three years.	• Training accuracy (93%) • Validation accuracy (94%).	which can provide 28% of Switzerland's annual electricity consumption. • The available roof area for PV installations in urban areas is 328 km², and 81% of the total ground floor area of each building can be used as the available roof area for PV installation. • The use of nanomaterial (graphene) in solar panels increases solar harvesting capacity by 40% compared to silicon solar cells. • The solar tracking system, developed with machine learning, maximizes energy conversion efficiency by following the Sun's path.	(Vatti et al., 2020)
Support vector machine (SVM) classifier	Classification of roof shapes to assess solar PV potential in buildings in the city of Geneva	Building data from the city of Geneva, Switzerland, including roof shapes and solar radiation data	Classification accuracy	• The SVM classifier was able to identify 6 types of roof shapes with an average accuracy of 66%. • Flat and complex roofs were classified more accurately than other types. Flat & shed roofs had the highest received mean solar radiation (809 kWh/m²), followed by cross/corner (806 kWh/m²) and hip (800 kWh/m²). • Complex roofs had the highest average PV potential, despite having the second-lowest received mean solar radiation, due to having the largest cumulative roof area and available or useful roof area for PVs.	(Mohajer et al., 2018)

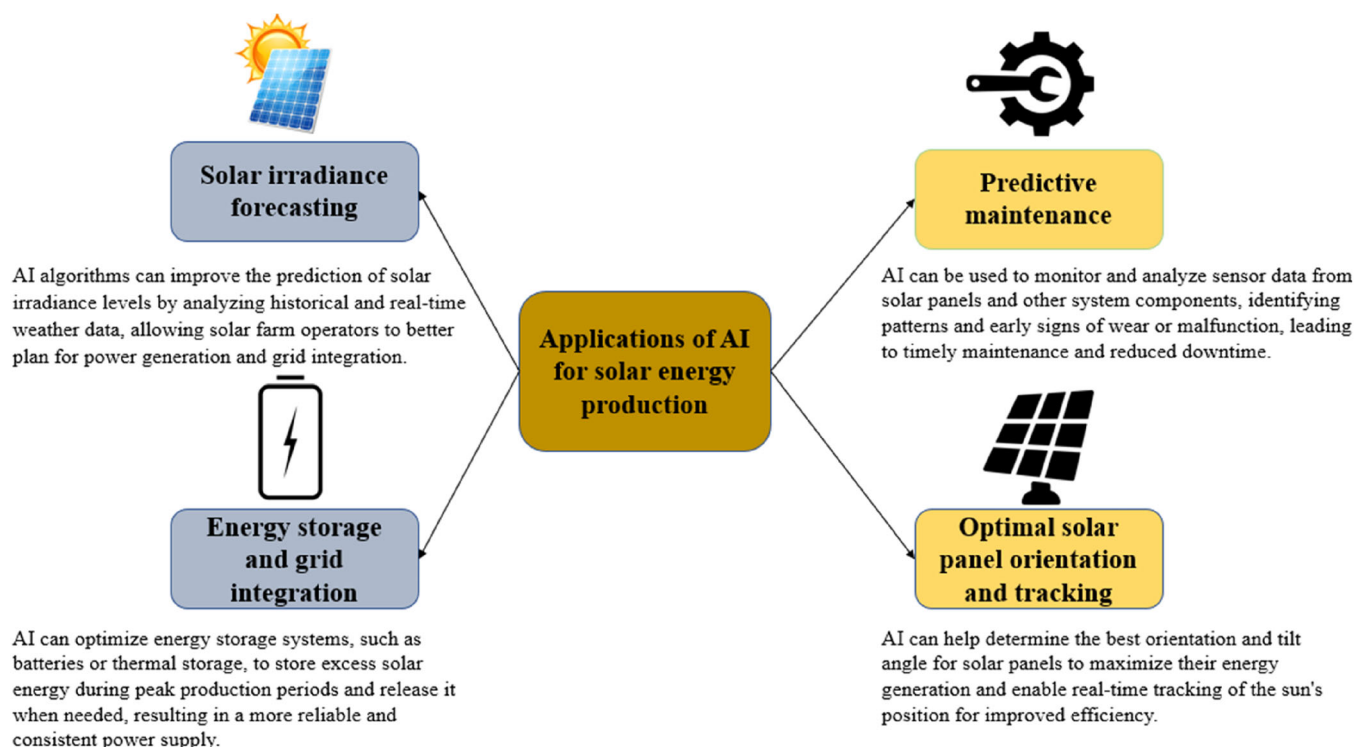


FIGURE 3 The application of artificial intelligence for solar energy production.

combustion fuels accounted for 48.7% (Elshafei et al., 2021). The Lake Nasser initiative was initiated to reduce Europe's reliance on fossil fuels and increase solar energy production. It is anticipated that 1000 km² of the lake will be covered with photovoltaic panels and that at least 30% of combustible fuel will be replaced in the first ten years, with total replacement expected by 2050. Elshafei et al. (2021) estimate that wind power, Photovoltaic, and Concentrated Solar Power will cost €50/MWh, €48/MWh, and €56/MWh in 2050, assuming a generation capacity of 2000 TWh/y. Despite differences in industrial and residential electricity costs, solar energy remains a more affordable option (Couture & Gagnon, 2010).

Since 2009, the price of solar panels has decreased by 80%, while their efficacy has increased, resulting in lower electricity costs (Couture & Gagnon, 2010). Solar energy technologies, specifically photovoltaics (PV), are expanding swiftly and are expected to play a significant role in electricity production worldwide. Between 2010 and 2015, solar PV prices decreased from \$0.285 per Kilowatt-hour to \$0.126 per Kilowatt-hour (Ferreira et al., 2018; Hoppmann et al., 2014). Additionally, as onshore wind and solar energy technologies become less expensive, a growing number of individuals and organizations are opting for renewable energy sources, which are more financially viable than other energy sources due to their low initial cost (Holm, 2013).

According to Ferreira et al. (2018), solar energy could produce 22% of the world's electricity by 2050, reducing carbon emissions. Solar energy technology and solar panel prices must be improved to make large-scale investments economically feasible. The cost of photovoltaic energy must be comparable to that of fossil fuels. Policy

support may be necessary until electricity prices incorporate environmental considerations and compete with fossil fuels. The declining cost of solar energy technology, particularly photovoltaics, has increased the number of individuals and organizations utilizing renewable energy sources. Some of the relevant organizations include the International Solar Energy Society (ISES), the Solar Energy Industries Association (SEIA), and the American Council On Renewable Energy (ACORE) (Singh, 2021). Solar energy remains an affordable alternative to conventional energy sources due to its low initial cost. However, additional research and development is required to make solar energy competitive with fossil fuels, and regulatory support may be necessary until power pricing incorporates environmental factors.

A study by Al-Nassar et al. (2019) delves into the economic feasibility of offshore wind energy in Kuwait, focusing on selecting optimal locations for windmill installations. The unique methodology combines meteorological data and the application of Weibull distribution parameters to assess wind power density at specific offshore locations. The study found Beacon N6, 3 km south of Boubiyan Island, to be the best marine location, having the highest wind power density, especially during summer when energy demand peaks. An intriguing outcome was that power production from offshore wind turbines was 30% cheaper per kWh than oil-based thermal power plants. A comprehensive financial analysis, utilizing a spreadsheet model on MS-EXCEL, was performed based on numerous economic factors, including interest rates, inflation, tax regulations, and capital expenditure. The research underscores the cost competitiveness of offshore wind energy in Kuwait, adding valuable insights to the global discourse on renewable energy economics (Al-Nassar et al., 2019).

TABLE 2 The various applications of machine learning to wind energy.

Machine learning technique	Application	Dataset	Performance metrics	Main results	Reference(s)
Random forest (RF) regression	Wind power forecasting based on daily wind speed data	Hourly wind speed observation values for 5 years from Nigde, Turkey, and four other datasets generated for different regions of Turkey (Cesme, Mamak, Bozcaada, and Silivri). The dataset contains 43,800 observations in total.	R ² values Mean absolute error (MAE) Root mean squared error (RMSE)	RF algorithm showed the best performance with an R ² value of 0.995 and MAE of 7.048.	(Demolli et al., 2019)
Artificial neural networks (ANN)	Wind energy production prediction	Data from open power system data (2020) Two data sources aggregated, containing weather and energy output information of wind turbines	Coefficient of determination (R2) Mean absolute error (MAE) Root mean squared error (RMSE)	ANN had the best performance with the highest R2, lowest MAE, and lowest RMSE, but had a longer training time and required more computational power.	(Buturache & Stancu, 2021)
Neural network regression (using TensorFlow and keras libraries)	Prediction of power generated by airborne wind energy (AWE) systems.	Aggregated data from seven tests consisting of numerical and categorical data from multiple sensors.	Mean squared error (MSE).	- Neural network model was successful in predicting tether force for AWE systems without hyper-parameter tuning. - Neural network model had a longer training time than other algorithms but had better overall accuracy performance.	(Rushdi et al., 2020)
Wavelet decomposition method, long short-term memory (LSTM) neural network, and multilayer perceptron (MLP)	Identifying suitable locations for offshore wind TurbineGenerators (OWTG) installations in the mediterranean basin and predicting short-term wind speed	Modern-era retrospective analysis for research and applications version 2 (MERRA-2) re-analysis dataset	Mean absolute percentage error (MAPE), Root mean square errors (RMSE), and Mean absolute error (MAE)	Identified several mediterranean basin zones with attractive offshore wind potential, such as the Aegean Sea, Gulf of Lyon, Northern Morocco, and Tunisia regions. The combined forecasting model (WTD-LSTM) demonstrated a higher level of accuracy in predicting short-term wind speed compared to single models (MLP and LSTM).	(Majidi Nezhad et al., 2022)
Support vector machine (SVM) classification	Analysis and optimization of photovoltaic (PV) system performance	Monitoring data of a 270 W mono-crystalline PV module over seven months, including I-V curves, electrical parameters, temperature, and weather parameters	Temperature coefficient of photovoltaic modules	The SVM classification method allowed for efficient targeting of common operating conditions and isolation of marginal phenomena in PV system performance analysis	(Hafdaoui et al., 2022)

(Continues)

TABLE 2 (Continued)

Machine learning technique	Application	Dataset	Performance metrics	Main results	Reference(s)
Support vector regression (SVR)	Energy management protocol for efficient and reliable operation, scheduling, maintenance, and energy trading in the modern power system using wind power and electric vehicles (EVs).	Wind measurements gathered in the framework of a wind power assessment project in Utique, Bizerte, Tunisia. One year of 10-minute measurements of wind speed, temperature, pressure, humidity, and wind direction at different heights.	Root squared error (RSE)	SVR model showed excellent capability in wind speed forecasting with an RSE of 0.94.	(Khabbouchi et al., 2023)
Deep reinforcement learning (DRL)	Wind turbine rotor speed optimization considering rain erosion	Rain intensity and wind speed distributions Physics-based simulation results for WTB coating fatigue life	WTB blade coating fatigue life extension Reduction in energy yield	Proposed method extends the predicted WTB blade coating fatigue life by 2.55 times. Minor reduction in energy yield (0.027%) compared to the original rotor speed schedule. Significantly reduced computational time compared to traditional gradient and evolutionary design optimization methods.	(Fang et al., 2022)
Deep deterministic policy gradient (DDPG) algorithm					
Kriging surrogate model					
Wavelet transform (WT)	24-h ahead wind power prediction (WPP) for a wind generation system in Beijing, China	Two-year (2014–2015) historical dataset of meteorological forecast, wind speed, and power real observations 85% of the dataset is used for model training, and 15% is used for validation Forecast performance is tested using a one-year (2016) testing dataset	Mean absolute percentage error (MAPE) Normalized mean absolute error (NMAE) Variance of forecast error Forecast skill (FS)	The WT-ACO-FFANN ML method improves 24-h ahead wind power forecast and outperforms other four evaluated methods (persistence, dual-step BP-FFANN, dual-step GA-FFANN, and dual-step PSO-FFANN) in MAPE, NMAE, variance, and FS. Daily mean MAPE, NMAE, and FS values achieved are 4.67%, 0.82%, and 56.22%. Average computation time to generate 24-h ahead forecasts is about 10 s using MATLAB	(Ma & Zhai, 2019)

TABLE 2 (Continued)

Machine learning technique	Application	Dataset	Performance metrics	Main results	Reference(s)
Multilayer perceptron (MLP) Radial basis function (RBF)	Wind power prediction for ducted wind turbines (DWTs)	Database on ducted wind turbines with 9000 data sources Input data: free stream wind speed, wind speed at the throat section, flow intensity in throat section, and the available power at the inlet section of the duct Output data: generated power of ducted wind turbines	Mean squared error (MSE) Coefficient of determination (R2)	software on a computer with Intel core i5-5200CPU, 2.20 GHz processor, and 4 GB RAM. For the MLP method, the best prediction performance (R2) of 0.99949 and MSE of 1.034e-4 at 77 epochs was achieved using the Bayesian Regularization backpropagation method with a tangent sigmoid transfer function in hidden layers and a linear transfer function in the output layer. For the RBF method, the best prediction performance (R2) of 0.99827 and MSE of 0.000353 at 70 epochs was achieved using a Gaussian function.	(Taghinezhad & Sheidaei, 2022)
Support vector machine (SVM) and radial basis function kernel (RBFK)	Imbalance classification of a scaled-down wind turbine using vibration signals	Vibration signals acquired under certain work conditions, grouped into 3 or 7 categories; same dataset used in previous works for comparison	The success rate of classification	For three-class classification, the best average success rates were 99.96% (RBFK) and 98.73% (SVM) at 900 rpm For seven-class classification, the best success rate was 98.83% (RBFK) at 900 rpm RBFK classifier had superior performance compared to other methods at 1200 rpm rotation speed	(de Oliveira Nogueira et al., 2022)
Multi-agent deep reinforcement learning framework Value-decomposition network (VDN) for multiple agent cooperation optimization	Optimizing energy management of building energy systems (BES) with the integration of renewable energy sources	Simulation of a building energy system with a period of 3 days, divided into 1440 periods (1-min time steps)	Uncomfortable duration (minutes) Unconsumed amount of renewable energy (kWh) Energy cost (CNY)	Multi-agent cooperation algorithm reduced the uncomfortable duration by 84%, the unconsumed amount of renewable energy by 43%, and the energy cost by 8% compared to the conventional rule-based control approach.	(Shen et al., 2022)

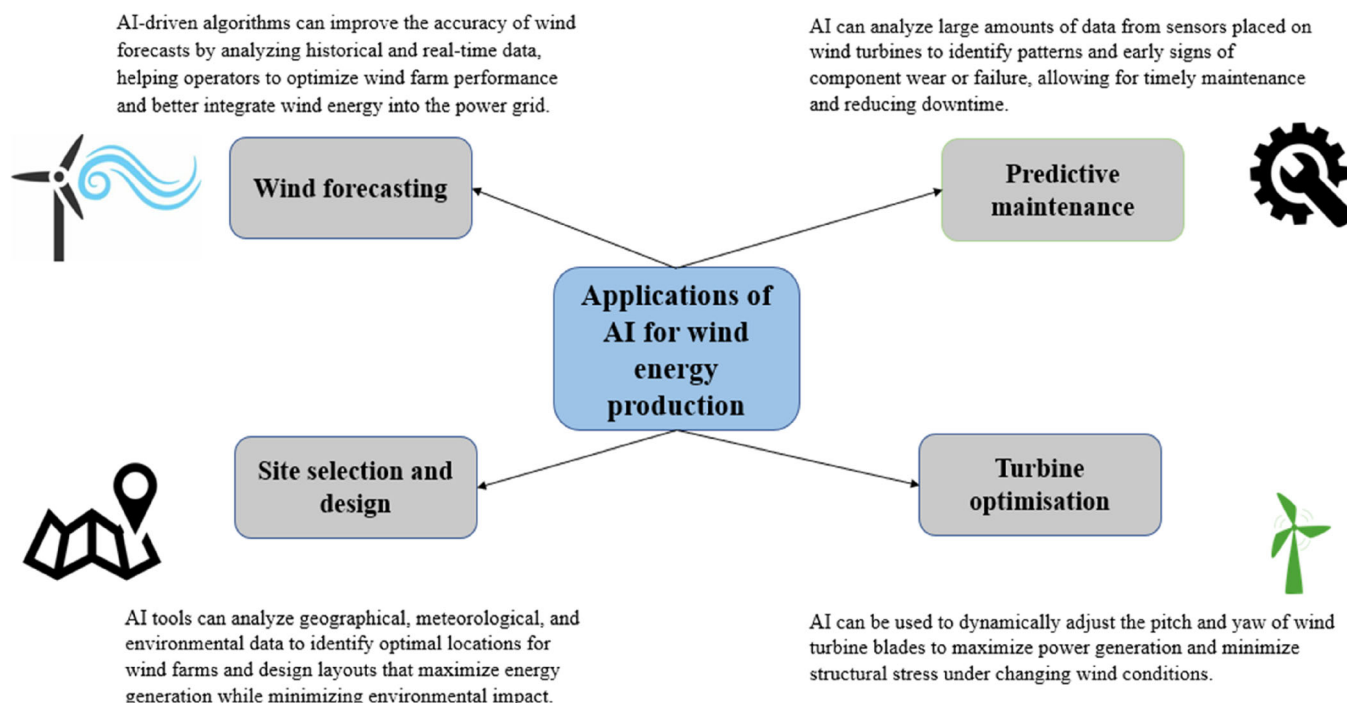


FIGURE 4 The applications of artificial intelligence for solar energy production.

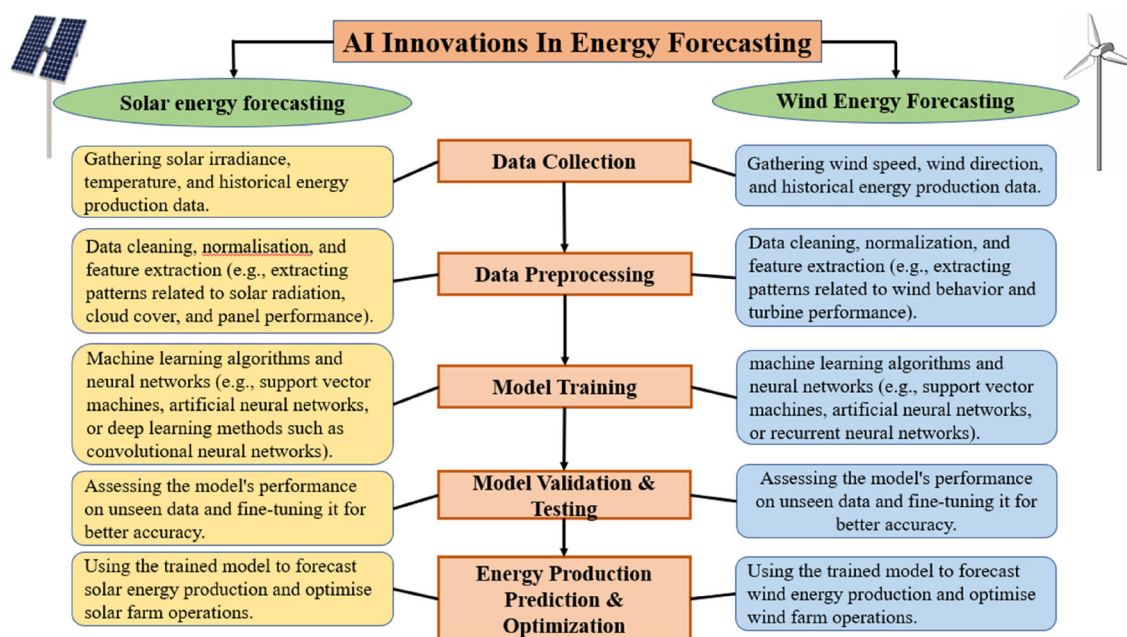


FIGURE 5 The AI innovations in energy forecasting steps.

The research by Castro-Santos et al. (2020) evaluated the economic viability of floating offshore wind farms along Portugal's continental coast using a multi-phased methodology that considered various factors, including offshore wind characteristics, wave height and period, bathymetry, and infrastructure proximity. The study assessed three types of wind platforms: semisubmersible, TLP, and spar, as well as two electricity rate scenarios. The spar platform had the lowest Levelized Cost Of Energy (LCOE) at 289.49 €/MWh,

followed by the semisubmersible platform (303.97 €/MWh) and the TLP platform (325.64 €/MWh). Internal Rate of Return (IRR) and Net Present Value (NPV) analyses indicated economic viability for spar and semisubmersible platforms only at the higher tariff. Near the Galician border, Northern Portugal was determined to be the optimal installation site. This paper highlights the significance of cost reduction and the applicability of this technique to other regions considering offshore wind farms (Castro-Santos et al., 2020).

Bahrami et al. (2019) assess wind energy potential in Uzbekistan, which relies heavily on fossil fuels to generate electricity. Utilizing hourly wind speed data from 17 different sites, the study calculated performance metrics, including power density, energy production, capacity factor, and levelized cost of energy (LCOE). According to the research, the average annual wind speed ranges from 0.61 to 3.98 m/s, the power density ranges from 1.74 to 88.55 W/m², and the energy production ranges from 15.27 to 775.72 kWh/m². Nukus, Kungrad, Ak Bajtal, and Buhara were identified as the most promising locations for wind energy development. Comparisons of commercially available wind turbines indicated that the Goldwind GW 140/3.0 was the most practicable model, with the highest energy production and capacity factor (Bahrami et al., 2019).

Himri et al. (2020) examine the economic viability and potential of wind power in Algeria's southwest region. The researchers analyzed wind characteristics using hourly mean wind data from 2003 to 2008. They identified key factors, including mean wind speed, energy flux, dominant wind direction, frequency distribution, and Weibull distribution scale and shape. Using RETScreen, the energy output and economic viability of a proposed wind farm in the Adrar region were evaluated. With an annual energy production of 3146 MWh, a capacity factor of 36%, and a cost of electricity production lower than the wholesale price during the feed-in tariff period, the study determined the project to be economically viable. In addition, the research revealed the project's potential to prevent 35,525 tonnes of Greenhouse Gases (GHG) emissions over a quarter-century (Himri et al., 2020).

To conclude, although solar and wind energy hold great promise, continued research, technological advancements, and site-specific feasibility studies are required to ensure these resources compete with conventional energy sources. This could transform the energy landscape, leading to an economically and environmentally viable future.

5.2 | Environmental and social impacts

The pursuit of energy sustainability necessitates a thorough comprehension of the environmental impacts of diverse energy sources and technologies. According to Shahsavari and Akbari (2018), all energy sources contribute to pollution, with the amount of pollution produced by combustion power plants contingent on the fuel type and technology employed. For instance, photovoltaic systems' efficiency and environmental impact vary based on the materials and production methods employed. Shahsavari and Akbari (2018) also examine the complexities of photovoltaic technologies, their efficacy, and the measures necessary to minimize the environmental impacts of their production and use. In addition, it emphasizes the need for pure, affordable energy solutions in developing nations, where using wood and solid fuels for cooking and heating contributes significantly to environmental degradation and health problems. The potential of renewable energy sources like solar and wind power can be harnessed to provide clean, affordable electricity while minimizing environmental

impacts, paving the way for a sustainable energy future by focusing on design, monitoring, and mitigation measures.

For instance, hydrogenated second-generation a-Si TFPV technology offers higher absorption rates in energy efficiency than crystalline PV. Still, its bandgap is less adapted to the solar spectrum than crystalline PV's 1.12 eV bandgap. Light can induce Staebler-Wronski degradation in a-Si TFPV. Variable bandgap PSCs deposited at low temperatures have an efficiency of 24.2% but suffer from limited lifetimes, high Voc, water sensitivity, batch-to-batch quality control issues, and Pb and Sn toxicity. CZTS could supplant CIGS TFPV with a maximum cell efficiency of 12.6% in the laboratory. Organic photovoltaics (OPVs) attain a maximum cell efficiency of 15% in the laboratory. Still, they face challenges in exciton transport, long-term stability, large-area deposition yield, and final efficiency (Hayat et al., 2019).

Farrokhi Derakhshandeh et al. (2021) discuss various solar panel cleaning techniques, such as passive coating, electrostatic and mechanical self-cleaning, and surface acoustic waves. In sweltering conditions, spraying water on solar panels can improve efficiency, cell count, and equipment utilization. Since there are 1480 metric tonnes of discarded solar photovoltaic cells worldwide, recovering tellurium is crucial for economic and environmental reasons.

Approximately 2.7 billion people in Sub-Saharan Africa and emerging Asia rely on wood and other solid fuels for cooking and heating, posing health, environmental, and economic hazards (Farrokhi Derakhshandeh et al., 2021). In 2012, 4.3 million fatalities were associated with using kerosene and solid biomass for cooking, while wood fuel consumption contributed to deforestation in East Africa and South Asia. Although recent advances in the bioenergy sector, particularly in the realm of biofuel conversion, have been significantly enhanced by the integration of machine learning technologies, as explored in the work by Wang et al. (2022). There is still a need for a safer, economically viable, and sustainable energy generation method.

Solar and wind energy can provide a cost-effective solution to this problem. Although its production requires energy, complex equipment, and toxic chemicals and gases, solar energy can power rural areas, create employment, and reduce utility costs. Installations of solar power plants can impact soil, water, biodiversity, habitat quality, and habitat degradation. However, properly maintained solar plants can provide habitats for endangered plant and animal species and utilize marginal and degraded land. By avoiding sensitive areas, employing bird-friendly materials, and monitoring avian behavior, it is possible to prevent bird deaths at solar power plants. Design, monitoring, and mitigation can be used to address the environmental impacts of solar energy installations (Shahsavari & Akbari, 2018). Obtaining energy sustainability is crucial for mitigating environmental impacts, as is effectively managing contamination from all energy sources. Different PV technologies affect both production efficiency and environmental impacts. Cleaning solar panels and recycling PV detritus can aid in mitigating environmental impacts and enhancing production effectiveness.

Tazi et al. (2019) critically examine the environmental impact of onshore wind farms in France's Champagne-Ardenne region, focusing

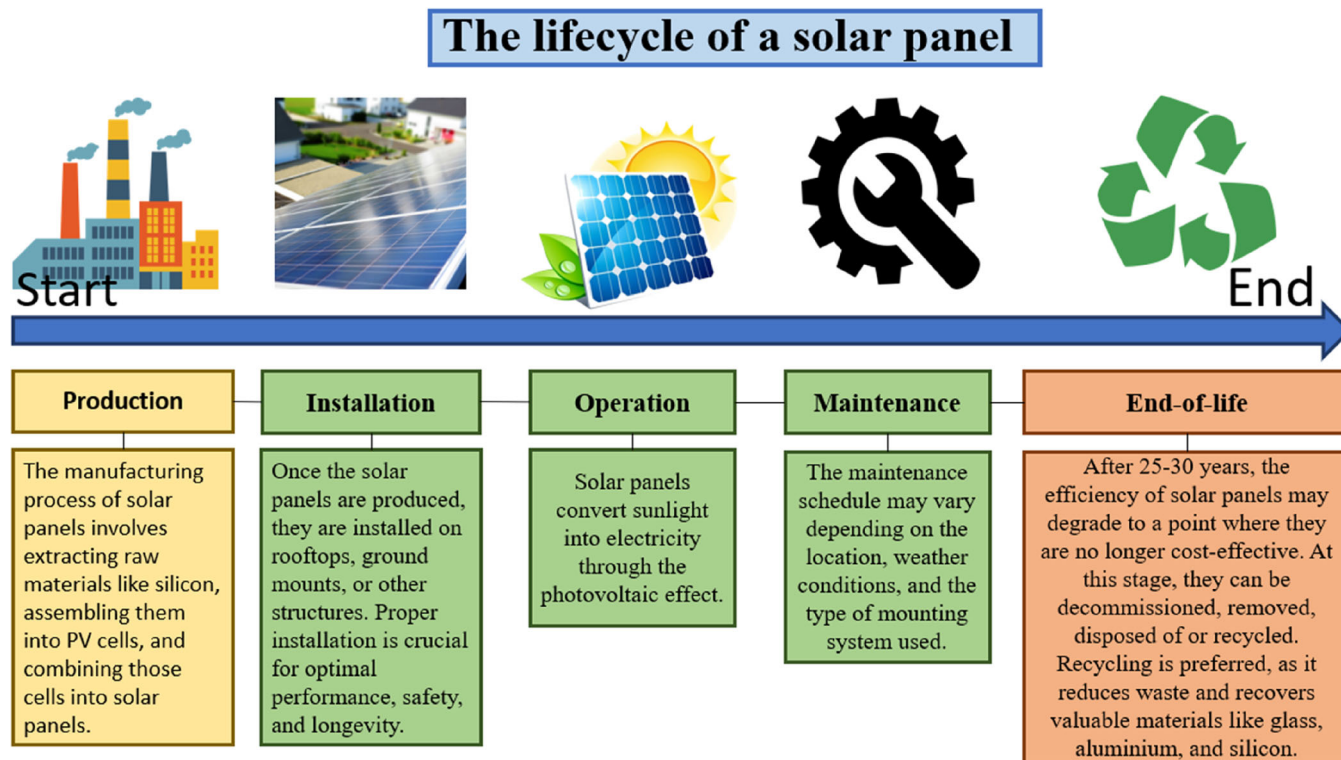


FIGURE 6 The schematic lifecycle of a solar panel.

on the waste management of wind turbines. The authors applied a Material Flow Analysis (MFA) model to quantify the waste from 2002 to 2020. The results revealed over 1 million tons of material, primarily ferrous and non-ferrous metals, polymers, glass, and concrete, would be generated by end-of-life (EoL) wind farms. Blades and composite materials accounting for over 27,000 tons need special management. Interestingly, without considering the concrete foundation, 73% of a wind turbine can be recycled (Tazi et al., 2019).

Dhar et al. (2020) confirm the lower overall environmental footprint of wind power compared to conventional energy sources but also highlight several potential negative impacts. These include noise pollution, visual esthetics, ecological disruption, electromagnetic interference, and potential local climate change. Wind turbines' largely aerodynamic and mechanical noise can be intrusive, particularly if installed near residential areas (Dhar et al., 2020). The study suggests mitigation measures, including proper insulation and strategically placing wind farms in noisy areas. Visual pollution is also a notable concern, with wind farms altering rural or natural landscapes. There's also the potential for 'NIMBY' (Not-In-My-Back-Yard) opposition from residents. The paper proposes community education and increasing public participation in planning and operation processes (Dhar et al., 2020).

Nazir et al. (2020) discuss the environmental impacts of wind energy, a growing alternative to fossil fuels. The paper identifies various ecological implications of wind energy facilities, such as noise pollution, visual disturbances, soil erosion, and deforestation, and proposes potential solutions to these challenges. It uses computational fluid dynamics (CFD) techniques to simulate wind turbine

designs and their impacts. (Nazir et al., 2020) suggest that turbines cause noise pollution, which can be categorized into aerodynamic and mechanical noise, and this can be mitigated through improved design. Visual impact is another concern that can be addressed using Geographic Information system-based assessments. The paper also addresses the shadow flicker, soil erosion, deforestation, and the wake effect of turbines (Nazir et al., 2020).

In summary, achieving energy sustainability requires thoroughly assessing environmental effects and formulating plans to address them. We have to further analyze the lifecycle of solar and wind energy using various methods, including lifecycle assessments. Figures 6 and 7 show the main process steps of the lifecycle of a solar panel and wind turbine. Solar and wind power are essential components of this pursuit, providing affordable and clean energy options. While these technologies also have environmental consequences, studies indicate they can be mitigated by implementing better designs, strategic placement, strong waste management, and community involvement. Ultimately, a blend of renewable energy technologies and thoughtful strategies could offer a sustainable way to meet future energy demands.

6 | ENERGY DEMAND PROJECTIONS, OPPORTUNITIES, AND CHALLENGES

As the global population and economy continue to expand, energy demand is anticipated to increase significantly over the next several

The lifecycle of a wind turbine

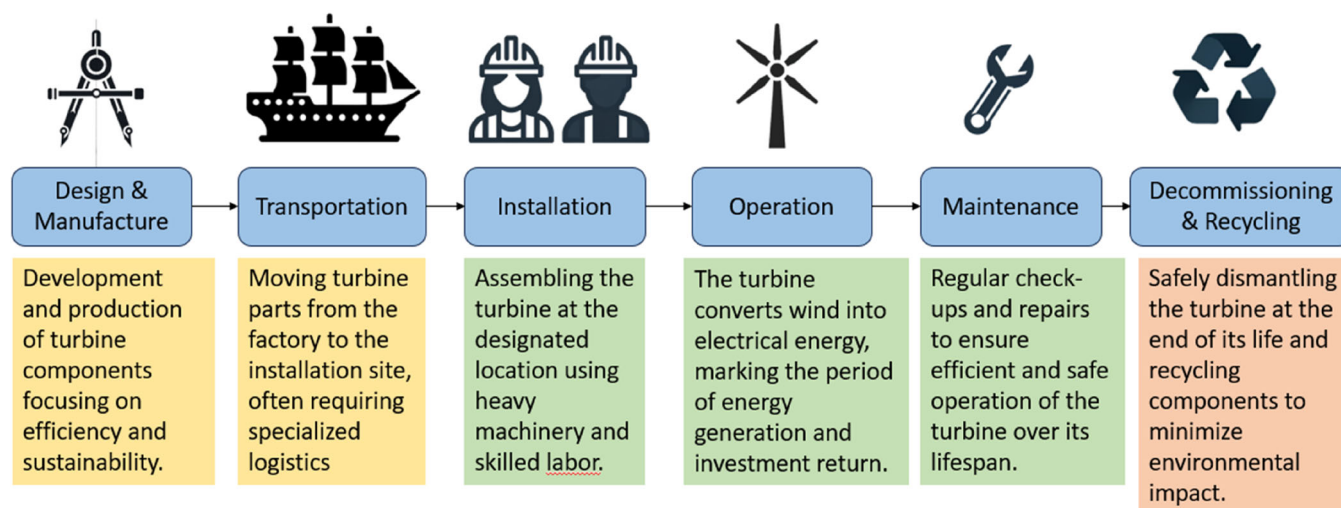


FIGURE 7 The schematic lifecycle of a wind turbine.

decades (Kober et al., 2020). This presents challenges and opportunities for the renewable energy industry, especially solar and wind energy solutions.

The intermittent nature of solar and wind power is one of the primary obstacles to reaching the growing energy demand. Solar and wind capacity have grown rapidly, but they must increase by a factor of seven between 2010 and 2050 if two-thirds of the world's energy consumption is to come from renewable sources (Lowe & Drummond, 2022). In addition, quicker implementation of bio jet fuels, bio-fuels for transportation, and industrial solar heat are necessary. This necessitates exploring innovative biofuel production methods, such as algae-based biofuels, to meet transportation and industrial energy demands sustainably. In addressing these obstacles, artificial intelligence can play a crucial role. AI-driven optimization techniques can increase the efficacy and cost-effectiveness of renewable energy systems, allowing them to compete with fossil fuel-based energy sources more effectively (Song et al., 2022). Specifically, machine learning algorithms can predict energy output from solar and wind sources, optimizing grid management and reducing reliance on backup fossil fuel sources. Integrating intermittent renewable energy sources with flexible dispatchable power, transmission connectivity, storage, smart infrastructure, and demand-side management can be facilitated by AI.

Solar and wind energy provide distinct opportunities beyond the production of electricity. Wind energy can reduce fossil fuel pollution (Pata et al., 2022; Sahu, 2018), whereas solar energy can be utilized for desalination, heating, and photocatalysis. Future research should also focus on enhancing wind turbine designs for greater efficiency, particularly in urban and low-wind environments. As these technologies continue to develop, their applications will likely expand, creating new opportunities for sustainable energy solutions. Emerging applications like solar-powered water purification systems could revolutionize access to clean water in remote areas. Integration of intermittent

renewable energy sources into the power grid requires energy storage. While lithium-ion batteries are the most prevalent energy storage technology, their environmental impact is a cause for concern (Pellow et al., 2020). Investigating solid-state batteries could offer a breakthrough in energy density and safety for renewable energy storage.

Alternative storage technologies, such as pumped hydro and compressed air energy storage, offer promising opportunities due to their low environmental impact despite their energy density and geographical limitations (Borri et al., 2020). Research into modular and scalable storage solutions, like flow batteries, could provide more versatile energy storage options. The low-cost, high-energy density and extended cycle life of latent and sensible heat storage devices have attracted attention as a potential solution for more sustainable energy storage. Developing advanced materials for thermal storage systems could significantly improve efficiency in harnessing and utilizing solar and wind energy. Future research should concentrate on improving the capacity and efficiency of renewable energy sources, especially solar and wind power, to meet the growing global energy demand. It should also prioritize addressing intermittency, a significant obstacle for these renewable energy sources (Richards et al., 2012; Sánchez et al., 2022).

Innovative solutions like hybrid renewable energy systems, combining solar, wind, and bioenergy, can provide a more stable and continuous power supply. In addition, the rapid development and deployment of energy storage technologies, particularly those with a smaller environmental footprint, are crucial (Parra et al., 2017). Particular emphasis should be placed on these storage technologies' life-cycle analysis and recyclability to ensure environmental sustainability. In addition, research into AI-driven optimization techniques that boost the performance and cost-effectiveness of renewable energy systems, allowing them to compete more effectively with fossil fuels, will be essential (Hassan et al., 2023). The expanded applications of wind and

solar energy beyond the generation of electricity, such as desalination, heating, and photocatalysis, require additional research (Ding et al., 2021; Zhang et al., 2018). Integrating solar and wind energy in smart building designs can lead to more energy-efficient urban environments.

In conclusion, solar and wind energy solutions can meet the rising energy demand. Still, challenges are presented, such as the need for quicker capacity expansion, technological advancements, and overcoming the intermittent nature of these energy sources. However, numerous opportunities for innovation and expansion are offered. Artificial intelligence can be crucial in optimizing renewable energy systems, enhancing efficiency, and facilitating their incorporation into the power infrastructure. These opportunities must be embraced, and challenges must be addressed to advance closer to a sustainable and resilient energy future.

7 | CONCLUSION

Renewable energy sources, such as solar and wind, have the potential to offer clean and affordable energy to meet the world's energy requirements. With technological improvements, harvesting energy from these sources has become increasingly viable and economical. Although obstacles such as cost, energy storage, and integration with existing energy networks still exist, solar and wind energy are expected to become more prevalent soon. Machine learning has shown great potential in driving the widespread adoption and increasing the feasibility of solar and wind energy for a sustainable future. In wind energy, research and development needs include investigating the environmental and social impacts of wind turbines, designing and testing wind turbine components, developing innovative materials, enhancing energy management and storage systems, and exploring offshore wind technologies. Future studies on the design of wind turbine blades should concentrate on reducing prices, boosting energy output, examining alternate materials, and recycling used blades. In addition, there have been developments in offshore tower design and evaluation under extreme stress conditions. For a sustainable energy future, the growth and deployment of renewable energy must be supported by investments. Renewable energy, such as solar and wind energy sources, has enormous potential to assist in achieving sustainable development goals and mitigating the effects of climate change.

CONFLICT OF INTEREST STATEMENT

The authors declare no conflict of interest.

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